Design Specification — VBF-T3R1-DS-001

Cell Design Specification — VBF-T3R1-DS-001

Field	Value	Source
Case	exp/t3_r1 — power-tool battery design	log.jsonl entry 0
Document number	VBF-T3R1-DS-001	delivery index
Generation date	2026-08-25	—
Status	Final (verdict: achieved, log.jsonl final entry)	bda log-evaluate R4

1. Design requirements (entry-0 criteria, verbatim from task text)

Metric	Threshold	Layer
Nominal capacity	>= 2.0 Ah	stage2
5C discharge retention	>= 0.95	stage2
Power density	>= 4000.0 W/kg	stage2
Maximum temperature (4C charge @45 C)	<= 333.15 K (60.00 C)	stage3
Lithium plating at 4C charge	False (none)	stage3

2. System and chemistry

Item	Value	Source
Base parameter set	Chen2020 (NMC811 / graphite, 1M LiPF6 EC/EMC)	SKILL.md anchor table; task text names no electrode system -> default with record
Cathode	NMC811, 40 um, porosity 0.42, particle radius 2.0 um	cell/p_v12.json
Anode	Graphite, 52 um, porosity 0.35, particle radius 2.5 um	cell/p_v12.json
Voltage window	2.5 - 4.2 V	Chen2020 set
Electrode area	0.065 m x 1.58 m = 0.1027 m2	Chen2020 set / calc-energy
Stack thickness	40+52+12+16+12 = 132 um	layer sum, calc-energy thickness_m

3. Cell architecture (final design V12_BalancedCooling)

Layer	Thickness [um]	Porosity	Particle radius [um]	Density [kg/m3]	Mass [g]	Source
Positive electrode (NMC811)	40	0.42	2.0	3699	7.77	Chen2020 set

Negative electrode (graphite)	52	0.35	2.5	2060	5.75	Chen2020 set
Separator (polyolefin)	12	0.47	-	1548	0.26	Chen2020 set
Positive current collector (Al)	16	-	-	2702	4.44	Chen2020 set
Negative current collector (Cu)	12	-	-	8933	11.04	Chen2020 set
Total (electrolyte excluded by contract)	—	—	—	—	29.26	calc-energy

Layer masses are mechanically computed: mass = layer_kg_m2 x area (calc-energy output r4_v12_energy.json).

4. Electrolyte design

Property	Value	Baseline (Chen2020)	Source
Conductivity kappa	1.5 S/m	0.9487	cell/p_v12.json
Diffusivity D	2.5e-10 m2/s	1.769e-10	cell/p_v12.json
Cation transference t+	0.35	0.2594	cell/p_v12.json

Rationale (round-2/3 attribution, log.jsonl): high-transport electrolyte is load-bearing for both 5C retention

and plating margin — V9 (kappa 1.2 / D 2.0e-10 / t+ 0.30) passed retention but eroded the plating margin to

+0.6 mV (cell/r3_v9_4c.json), so the aggressive values are retained. Values are transport-parameter

overrides of the baseline formulation (electrolyte formulation degree of freedom, widest interpretation

recorded in entry-0 meta.freedoms).

5. Thermal design

Property	Value	Source
Cooling coefficient h	20 W/m2/K	cell/p_v12.json
Ambient for rating	25 C (discharge), 45 C (4C charge protocol)	bda protocols

Rationale (round-3/4 attribution): h trades thermal margin against plating margin (warmer cell kinetically

suppresses plating): h=15 -> T_max 331.96 K / anode +14.9 mV (V8); h=30 -> T_max 326.86 K / anode +8.2 mV

(V11). h=20 lands both margins balanced (measured: T_max 329.77 K, anode min 12.1 mV)

and is realistic for a power tool (natural convection + tool-body conduction with light airflow).

6. Electrical ratings

Rating	Value	Source

Nominal capacity Q_nom	3.08 Ah	cell/p_v12.json (fixed-point iteration: measured 1C = 3.075 Ah, round 2 lesson)
1C current	3.08 A	Q_nom x 1
5C discharge current	15.4 A	Q_nom x 5
4C charge current	12.32 A	Q_nom x 4
Midpoint voltage	3.844 V	calc-energy midpoint_voltage_v

7. Verification summary

See DVPR (VBF-T3R1-DVPR-001): all five criteria verified pass by DFN simulation, evidence cell/r4_v12_*.json via bda log-evaluate (round 4, 5/5 checked, verdict pass). Iteration history in log.jsonl (rounds 1-4) and report.html.

8. Design notes (honest limits)

- Electrolyte mass excluded from calc-energy by contract (parameter set lacks density; output electrolyte_included: false) — mass/power density slightly optimistic vs a real cell with electrolyte.
- True DFT/MD endorsement skipped (real_compute=false, entry-0 meta): no first-principles values claimed.
- 4C CC-only charge accepts ~0.81 Ah before the 4.2 V ceiling (cell/r4_v12_4c.json) — a CV taper phase is required for full charge; plating-free result is for the CC segment per protocol definition.
- Manufacturability drawings with tolerances, material datasheets, and process cards are outside the pure-simulation boundary and are not provided.